

1 Solution — GIAM 2.3

Problem 2

1.1 Problem

Use truth tables to verify or disprove the following logical equivalences.

- (a) $(A \wedge B) \vee B \cong (A \vee B) \wedge B$
- (b) $A \wedge (B \vee \neg A) \cong A \wedge B$
- (c) $(A \wedge \neg B) \vee (\neg A \wedge \neg B) \cong (A \vee \neg B) \wedge (\neg A \vee \neg B)$
- (d) The absorption laws.

1.2 Answer

All four are valid logical equivalences.

Moreover each one collapses dramatically: (a) both sides equal B ; (b) both sides equal $A \wedge B$; (c) both sides equal $\neg B$; (d) both sides equal A .

1.3 The Truth Tables

All four hold — the two columns agree in every row, and each side collapses to something simple.

(a) $(A \wedge B) \vee B \models (A \vee B) \wedge B$			
A	B	$(A \wedge B) \vee B$	$(A \vee B) \wedge B$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	F	F
✓ VALID — both columns are just B			

(b) $A \wedge (B \vee \neg A) \models A \wedge B$			
A	B	$A \wedge (B \vee \neg A)$	$A \wedge B$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F
✓ VALID — the $\neg A$ disjunct is absorbed			

(c) $(A \wedge \neg B) \vee (\neg A \wedge \neg B) \models (A \vee \neg B) \wedge (\neg A \vee \neg B)$			
A	B	left side	right side
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	T
✓ VALID — both columns are just $\neg B$			

(d) absorption: $A \vee (A \wedge B) \models A$, $A \wedge (A \vee B) \models A$				
A	B	$A \vee (A \wedge B)$	$A \wedge (A \vee B)$	A
T	T	T	T	T
T	F	T	T	T
F	T	F	F	F
F	F	F	F	F
✓ VALID — both collapse to A; B is absorbed				

All four are valid. Notice how much each one simplifies:

(a) both sides = B (b) both sides = $A \wedge B$ (c) both sides = $\neg B$ (d) both sides = A

In every case a variable that looks essential turns out to be redundant.

Figure 1.1: Four panels of truth tables over A and B. Panel (a): the columns for $(A \text{ and } B) \text{ or } B$ and for $(A \text{ or } B) \text{ and } B$ are T, F, T, F — identical, and both equal B; valid. Panel (b): $A \text{ and } (B \text{ or not-}A)$ versus $A \text{ and } B$ give T, F, F, F in both columns; valid, the not-A disjunct is absorbed. Panel three: $(A \text{ and not-}B) \text{ or } (\text{not-}A \text{ and not-}B)$ versus $(A \text{ or not-}B) \text{ and } (\text{not-}A \text{ or not-}B)$ give F, T, F, T in both columns, and both equal not-B; valid. Panel (d), the absorption laws: $A \text{ or } (A \text{ and } B)$ and $A \text{ and } (A \text{ or } B)$ both give T, T, F, F, matching the column for A itself; valid, B is absorbed in each. Summary: all four valid, with (a) equal to B, (b) equal to A and B, the third equal to not-B, and (d) equal to A — in every case a variable that looks essential turns out to be redundant.

1.3.1 (a) $(A \wedge B) \vee B \cong (A \vee B) \wedge B$ — valid

A	B	$A \wedge B$	$(A \wedge B) \vee B$	$A \vee B$	$(A \vee B) \wedge B$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	T	T

F	F	F	F	F	F
---	---	---	----------	---	----------

Table 1.1

The two final columns match. Both are simply the column for B .

1.3.2 (b) $A \wedge (B \vee \neg A) \cong A \wedge B$ — valid

A	B	$\neg A$	$B \vee \neg A$	$A \wedge (B \vee \neg A)$	$A \wedge B$
T	T	F	T	T	T
T	F	F	F	F	F
F	T	T	T	F	F
F	F	T	T	F	F

Table 1.2

1.3.3 (c) $(A \wedge \neg B) \vee (\neg A \wedge \neg B) \cong (A \vee \neg B) \wedge (\neg A \vee \neg B)$ — valid

A	B	$A \wedge \neg B$	$\neg A \wedge \neg B$	left side	$A \vee \neg B$	$\neg A \vee \neg B$	right side
T	T	F	F	F	T	F	F
T	F	T	F	T	T	T	T
F	T	F	F	F	F	T	F
F	F	F	T	T	T	T	T

Table 1.3

Both sides reduce to $\neg B$.

1.3.4 (d) The absorption laws — valid

The two absorption laws are

$$A \vee (A \wedge B) \cong A, \quad A \wedge (A \vee B) \cong A.$$

A	B	$A \wedge B$	$A \vee (A \wedge B)$	$A \vee B$	$A \wedge (A \vee B)$	A
T	T	T	T	T	T	T
T	F	F	T	T	T	T
F	T	F	F	T	F	F
F	F	F	F	F	F	F

Table 1.4

Both compound columns match the A column exactly. ■

1.4 Remark — Part (a) *Is* the Absorption Laws in Disguise

Part (a) looks like a new fact, but it is really parts (d) applied twice. Reading each side with B as the “absorbing” variable:

$$(A \wedge B) \vee B = B \vee (B \wedge A) \stackrel{\text{absorption}}{=} B, \quad (A \vee B) \wedge B = B \wedge (B \vee A) \stackrel{\text{absorption}}{=} B.$$

Both sides independently collapse to B , so of course they are equal to each other. This is a useful habit: rather than comparing two messy expressions row by row, **simplify each side separately** and see whether they land on the same normal form. Here the shared normal form is the single letter B .

1.5 Remark — Part (b): Absorption in Its Complemented Form

Part (b) is the *complemented* cousin of absorption, sometimes listed separately as

$$A \wedge (\neg A \vee B) \cong A \wedge B, \quad A \vee (\neg A \wedge B) \cong A \vee B.$$

The algebraic derivation is a distribution followed by a cancellation:

$$A \wedge (B \vee \neg A) = (A \wedge B) \vee (A \wedge \neg A) = (A \wedge B) \vee \mathbf{F} = A \wedge B.$$

The point is that $A \wedge \neg A$ is a **contradiction** — always false — so that disjunct contributes nothing and drops out. Intuitively: once you have already assumed A , being told " B or not- A " gives you no new escape route, since “not- A ” is already ruled out; all you learn is B .

1.6 Remark — Part (c) and the Two Normal Forms

Part (c) is a nice illustration of the two standard **normal forms**. The left side,

$$(A \wedge \neg B) \vee (\neg A \wedge \neg B),$$

is in **disjunctive normal form** (an OR of ANDs), while the right side,

$$(A \vee \neg B) \wedge (\neg A \vee \neg B),$$

is in **conjunctive normal form** (an AND of ORs). Both simplify to $\neg B$ by factoring out the common piece and applying a tautology/contradiction:

$$\begin{aligned} \text{left} &= \neg B \wedge (A \vee \neg A) = \neg B \wedge \mathbf{T} = \neg B, \\ \text{right} &= \neg B \vee (A \wedge \neg A) = \neg B \vee \mathbf{F} = \neg B. \end{aligned}$$

Any formula can be written in *either* normal form, so the same truth function routinely wears two very different-looking syntactic costumes. Recognising that is precisely what these exercises train — and it is why SAT solvers insist on converting everything to one fixed form (CNF) before starting work.

1.7 Remark — Why Absorption Deserves Its Name

The absorption laws say that when a variable appears both alone and inside a nested clause, the nested clause is **absorbed** — swallowed with no trace:

$$A \vee (A \wedge B) = A, \quad A \wedge (A \vee B) = A.$$

The reasoning is one line each. For the first: if A is true the whole disjunction is true; if A is false then $A \wedge B$ is also false, so the disjunction is false. Either way the value is exactly A , and B never matters. Absorption has no analogue in ordinary arithmetic — $a + ab \neq a$ in general — which is one more sign (following Problem 1) that Boolean algebra is genuinely its own structure rather than arithmetic with new symbols.

1.8 Remark — Simplification Is the Practical Payoff

Every part of this problem ends with an expression shrinking: a four-symbol formula becomes a single letter. That is not a curiosity but the entire economic motive for studying these laws. In digital design (Problem 2.1.1), each connective is a physical gate costing area, power, and delay; discovering that $(A \wedge B) \vee B$ is just B means deleting two gates and a wire. The same is true in software — a compiler simplifying a Boolean condition, or a database engine simplifying a `WHERE` clause, is running exactly these identities. Truth tables *verify* an equivalence; the algebraic laws are what let you *find* one in the first place, and at scale the laws are the only practical option, since a truth table with n variables has 2^n rows.